Exercise Sheet 3

Generative Models and Naive Bayes

Probabilistic Machine Learning

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3.A The Generative Route

Exercise 3.A.1: Counting Parameters

A fraud classifier (two classes) uses four attributes: **payment** (card, invoice, wallet), **country** (domestic, EU, other), **new device** (yes/no) and **night** (yes/no).

1. How many free numbers does the full generative model need — the class prior and one joint $p(x \mid y)$ per class?
2. How many does naive Bayes need?
3. For p yes/no attributes and K classes, give both counts as formulas.
4. From which p on does the full joint need more than a million numbers *per class*?

3.B Naive Bayes by Hand

Exercise 3.B.1: Twelve Transactions

Flags: new device D, foreign IP F, night N, high amount H (1 = yes).

#	D	F	N	H	fraud?	#	D	F	N	H	fraud?
1	1	1	1	1	yes	7	1	0	0	0	no
2	1	1	0	1	yes	8	0	0	0	0	no
3	1	0	1	0	yes	9	0	1	0	0	no
4	0	1	1	1	yes	10	0	0	0	0	no
5	0	0	0	0	no	11	1	0	1	0	no
6	0	0	1	0	no	12	0	0	0	0	no

1. Estimate $P(\text{fraud})$ and $P(\text{flag} = 1 \mid \text{class})$ for all four flags by counting.
2. Compute $P(\text{fraud} \mid x)$ for $x_1 = (D, F, N, H) = (1, 0, 0, 0)$.
3. Do the same for $x_2 = (0, 0, 0, 1)$. What goes wrong, and why?
4. Repeat 2 and 3 with Laplace smoothing on the flags (keep the prior unsmoothed). Which result changes a lot, which only a little?

3.C Continuous Attributes

Exercise 3.C.1: Gaussian Naive Bayes by Hand

Twelve transactions: the amount as $x = \log_{10}$ amount in € ($x = 2$ is 100 €) and the flag new device D.

	x	$D = 1$
4 frauds	2, 2, 3, 3	3 of 4
8 honest	1, 1, 1, 1, 2, 2, 2, 2	2 of 8

1. Fit a normal per class to x: mean and standard deviation (divide by n).
2. A new transaction of 316 € ($x = 2.5$), device unknown. Compute $p(x \mid \text{fraud})$, $p(x \mid \text{honest})$, their ratio and $P(\text{fraud} \mid x)$ with the prior of the table.
3. It came from a new device. Update the posterior with Gaussian naive Bayes.
4. Would the likelihood ratio change if the densities were written over the amount in € instead of $\log_{10}$ amount? Why (not)?

Exercise 3.C.2: A Shared Spread Gives a Threshold

Both classes of 3.C.1 have the same spread, $\sigma = 0.5$.

1. Show that with equal spreads $\log \frac{p(x|\text{fraud})}{p(x|\text{honest})}$ is linear in x, and give it for the fitted numbers.
2. With the prior of the table, above which amount does the Bayes decision rule decide fraud?
3. The same model at the real base rate $1:999$: where is the cut now?
4. Write $P(\text{fraud} \mid x)$ as $1/(1+e^{-(wx+b)})$ for the prior of the table. Which model of the next lecture has this form?

3.D Deciding on a Class

Exercise 3.D.1: Costs, Prior and One Threshold

A naive Bayes fraud model at the base rate $1:999$. Checking a transaction costs €5 if it is honest; a fraud let through costs €500.

1. Above which posterior should a transaction be checked? Above which likelihood ratio?
2. Three transactions have likelihood ratios 6, 84 and 1,140. Which are checked?
3. Which of the three does the Bayes decision rule (equal costs) call fraud?
4. A new market has a fraud rate of 1 in 100 and the same fraud patterns. What is the threshold on the likelihood ratio there — and must the model be refitted?